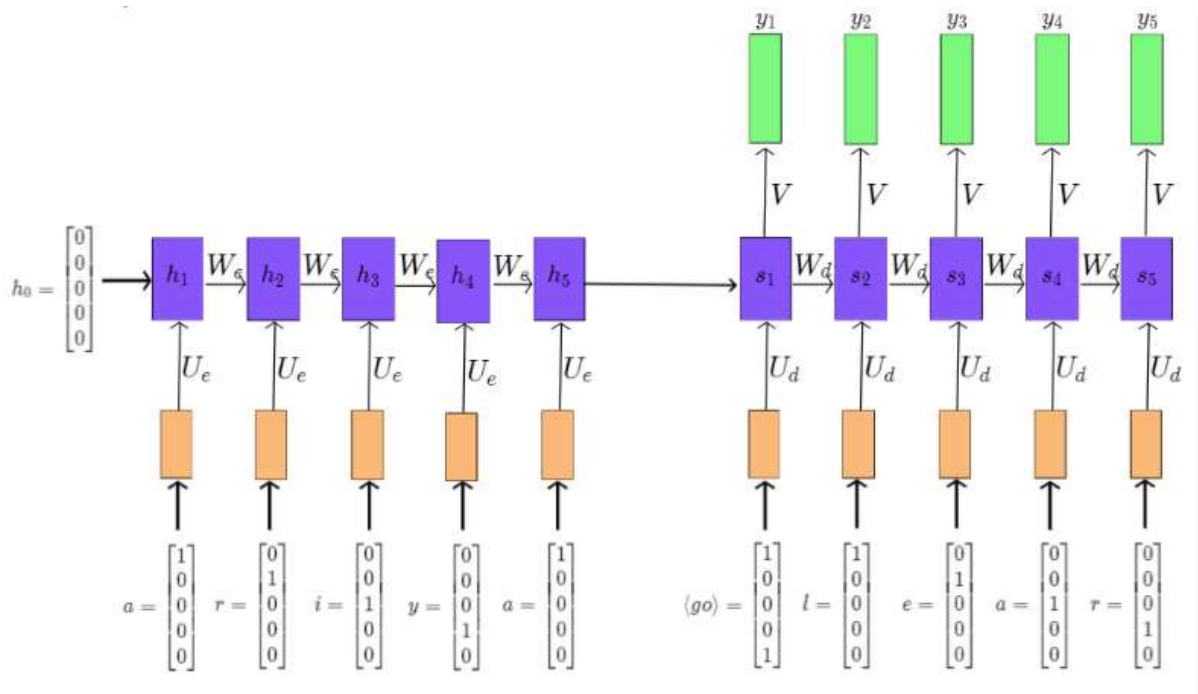


IITM BS Programming & Data Science

Deep Learning

Graded Assignments Solution : Week 11

In this assignment, we extend the RNN model implemented in the previous assignment to translate the sequence of characters " $x_{source} = \text{ariya}$ " (in Tamil) to " $x_{target} = \text{learn}$ " (in English) using a simple encoder decoder architecture as show below. Each character is represented using one hot encoding of size 5×1 as shown in the figure



The shape of the variables is as follows, $U_e, W_e, U_d, W_d, V \in \mathbb{R}^{5 \times 5}$

$h_t, s_t, x_{source}, x_{target} \in \mathbb{R}^{5 \times 1}$

y_t is a probability distribution over all 5 characters (in the vocabulary) at time step t . Let the initial state of the decoder is $s_0 = h_5$. Use cross entropy loss and gradient descent with $\eta=1$.

1) Suppose that we use the $\tanh()$ activation function to compute the state vectors of encoder and decoder (h_t, s_t), respectively. Do forward propagation with the initialized parameters. What is the total loss value $L(\theta)$?

Assume the character "n" is encoded as $[0\ 0\ 0\ 0\ 1]^T$

Ans. 17.48

Explanation

Colab link :

https://colab.research.google.com/drive/1R6YczqkWuQ2yT55mnLxWvzLkmHPeX_QJ?usp=sharing

2) Initialize the decoder with, $s_0 = h_5$. Run the decoder in Auto-regressive mode. The source sequence is translated into which of the following sequences?

☐ rnnrn

☐ leenl

☒ rnlInn

☐ learn

Explanation

Colab file

3) Train the model for 20 epochs. What is the total loss value now?

Ans. 0.05

Explanation

Colab file

4) With the updated parameters, compute h_5 of encoder and initialize the decoder with $s_0 = h_5$. Run the decoder(with the updated parameters) in Auto-regressive mode. The source sequence is translated into which of the following sequences?

☐ rnnrn

☐ leenl

☐ rnlInn

☒ learn

Explanation

Colab file

5) Shuffle the characters in the source sequence and repeat Q4. What do you observe?

Ans. Output remains 'learn'

Explanation

Colab file

6) Suppose that we use attention mechanism in the encoder and decoder model by computing the context vector as follows,

$$c_t = \sum_{j=1}^T \alpha_{jt} h_j$$

what is the length of T?

☒ 5

☐ 4

☐ 10

☐ 8

Explanation

In the context of an attention mechanism in encoder-decoder models, the length of T represents the number of time steps or positions in the encoder's hidden states.

If the input length is 5, then T , which represents the number of time steps or positions in the encoder's hidden states, will also be 5. This means that there are 5 hidden states h_j in the encoder, corresponding to each time step j from 1 to 5.

